

Competing Stroke Etiologies and Outcomes After Breakthrough Ischemic Stroke on Oral Anticoagulants in Patients With Atrial Fibrillation

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Abstract

Background and Objectives

Patients with atrial fibrillation (AF) who experience an ischemic stroke despite oral anti-coagulation (OAC) face a high risk of recurrence and bleeding. We aimed to compare 90-day outcomes between patients with and without competing stroke etiologies after a breakthrough ischemic stroke while on OAC for AF.

Methods

ASPERA-R retrospectively enrolled adults (>18 years) with AF who experienced a breakthrough ischemic stroke, defined as either first or recurrent event, occurring on continuous OAC (last intake ≤48 hours), across 35 centers from February 2020 to February 2025. Patients' eligibility required imaging-confirmed ischemic stroke, baseline vessel imaging, and a standardized 90-day follow-up. Patients with inadequate anticoagulant dosing or poor adherence were excluded. Competing etiologies were defined as any additional mechanism coexisting with cardioembolism and plausibly contributing to the index stroke. The primary outcome was 90-day recurrent ischemic stroke; secondary outcomes included modified Rankin Scale shift, myocardial infarction, and vascular and all-cause death. Bleedings were also evaluated. We performed inverse probability weighting to balance baseline covariates and applied Cox or regression models to compare outcomes.

Results

Among 1,649 patients (mean age 78 years, 47.8% male), 24.3% had competing etiologies, most commonly large artery atherosclerosis (59.9%). Considering the weighted population, 90-day recurrent ischemic stroke occurred more often in those with competing etiologies (6.0% vs 2.7%; adjusted hazard ratio [aHR] 2.62, 95% CI 2.01–3.42; $p < 0.001$). Risk was greatest for large artery atherosclerosis and other determined causes. Competing etiologies also had worse disability distribution (adjusted odds ratio 1.30, 95% CI 1.08–1.55; $p = 0.005$) and higher all-cause mortality (aHR 1.58, 95% CI 1.18–2.12; $p = 0.002$). Moderate-to-severe bleeding was higher with competing etiologies (aHR 1.82, 95% CI 1.08–3.09; $p = 0.026$), whereas 24-hour hemorrhagic transformation was less frequent (adjusted risk difference –4.3%, 95% CI –7.8% to –0.7%; $p = 0.020$). Other outcomes did not differ.

Discussion

One in 4 patients with AF and breakthrough ischemic stroke on OAC had competing etiologies, most often large artery atherosclerosis. These patients showed higher risk of recurrent stroke, functional dependence, mortality, and bleeding, highlighting the need for development of individualized secondary prevention. Main study limitations include observational retrospective design and incomplete data on on-treatment anticoagulation quality.

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Supplementary Material

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Glossary

AF = atrial fibrillation; **aHR** = adjusted HR; **aOR** = adjusted odds ratio; **aRD** = adjusted risk difference; **ASPERA** = Advancing Knowledge in Ischemic Stroke Patients on Oral Anticoagulants; **CDU** = color Doppler ultrasonography; **CTA** = CT angiography; **DOAC** = direct oral anticoagulant; **DSA** = digital subtraction angiography; **HR** = hazard ratio; **HT** = hemorrhagic transformation; **ICH** = intracranial hemorrhage; **IPW** = inverse probability weighting; **IQR** = interquartile range; **MRA** = MR angiography; **mRS** = modified Rankin Scale; **NCCT** = noncontrast CT; **NIHSS** = NIH Stroke Scale; **OAC** = oral anticoagulation; **SHR** = subdistribution HR; **sICH** = symptomatic ICH; **SMD** = standardized mean difference; **TOAST** = Trial of Org 10172 in Acute Stroke Treatment; **VKA** = vitamin K antagonist.

Trial Registration Information

ClinicalTrials.gov NCT06823466.

Introduction

Atrial fibrillation (AF) substantially increases the risk of ischemic stroke, and oral anticoagulation (OAC), preferably with direct oral anticoagulants (DOACs), is the cornerstone of secondary prevention.¹⁻³ Nevertheless, a subset of patients experience ischemic stroke while on OAC and subsequently face a persistently elevated risk of recurrence and death.⁴⁻⁶ A recent patient-level meta-analysis estimated this residual risk at 1.5%–2.5% per year, underscoring the clinical relevance of this population.⁷

Breakthrough strokes are mechanistically heterogeneous. OAC failure may result from subtherapeutic exposure (underdosing, poor adherence, drug interactions, impaired absorption, or organ dysfunction), persistent cardioembolic sources (atrial enlargement, cardiomyopathy, or left atrial appendage thrombus), or, less commonly, pharmacologic resistance.^{8,9} In addition, noncardioembolic mechanisms, particularly large artery atherosclerosis and small vessel disease, frequently concur with cardioembolism in determining both the index stroke on OAC and recurrent events.^{10,11}

The current evidence on the prognostic impact of concomitant mechanisms coexisting with cardioembolism in patients with AF receiving OAC, collectively termed “competing etiologies,” remains limited. The multicenter Causes and Risk Factors of Cerebral Ischemic Events in Patients With Nonvalvular AF Treated With NOACs for Stroke Prevention (RENO) study showed that recurrent stroke after a breakthrough event on OAC was strongly associated with competing etiologies, along with low or off-label DOAC dosing, atrial enlargement, dyslipidemia, and higher CHA₂DS₂-VASc scores.¹⁰ The prospective Novel Oral Anticoagulants in Ischemic Stroke Patients (NOACISP)-LONGTERM registry similarly reported competing mechanisms in approximately 1 in 5 AF patients with ischemic stroke and identified large artery atherosclerosis as an independent predictor of recurrence and composite vascular outcomes.¹¹ However, RENO included patients with irregular or discontinued anticoagulation,¹⁰ while NOACISP-LONGTERM

did not restrict inclusion to breakthrough events on OACs and focused mainly on large artery disease.¹¹ As a result, the prognostic impact of competing etiologies remains insufficiently defined, with narrow outcome assessment and limited patient characterization. A more granular understanding could refine recurrence risk prediction and contribute to developing improved secondary prevention strategies beyond anticoagulation alone.

Hence, the aim of this study was to compare 90-day outcomes between patients with and without competing stroke etiologies in addition to AF after a first breakthrough ischemic stroke while on OAC.

Methods

Standard Protocol Approvals, Registrations, and Patient Consents

The Advancing Knowledge in Ischemic Stroke Patients on Oral Anticoagulants (ASPERA) study was approved by the Abruzzo regional ethics committee (CEtRA) in February 2025 (approval code 033054/25). Informed consent for being included in the ASPERA-R study was not required in most participating countries, where ethics committees granted a waiver because data were obtained solely through retrospective chart review. In jurisdictions where consent is mandated even for retrospective studies (i.e., Italy), written informed consent was obtained from patients or from a legally authorized representative; for deceased or unreachable patients, the ethics committee approved a waiver of consent. The study protocol was registered on ClinicalTrials.gov (unique identifier: NCT06823466).

Study Design

The ASPERA study is a large, multicenter, real-world observational investigation coordinated by the University of L'Aquila. It includes 2 cohorts: a retrospective arm (ASPERA-R), designed to assess baseline characteristics and 90-day outcomes after breakthrough ischemic stroke on any OAC for AF, and a prospective arm (ASPERA-P), aimed at evaluating long-term outcomes up to 5 years, which is still ongoing. In

this study, we present results from the ASPERA-R arm, based on data retrospectively collected from 35 stroke centers across 9 countries in Europe and North Africa (eAppendix 1). All patients were identified after experiencing the index breakthrough stroke, and all 90-day outcomes were extracted retrospectively from routine clinical follow-up records. The ASPERA-P arm is still ongoing. The data set for this analysis was finalized on September 1, 2025. Reporting followed the Strengthening the Reporting of Observational Studies in Epidemiology guidelines.¹²

Study Population

The ASPERA-R arm retrospectively enrolled adults who had a breakthrough ischemic stroke—defined as an ischemic stroke occurring despite ongoing, continuous OAC for AF, whether it was the first event or a recurrence—between February 2020 and February 2025. Continuous OAC use was defined as a documented prescription of uninterrupted intake during the 7 days preceding the index event, with the last dose taken within 48 hours before symptom onset. Patients with previous ischemic stroke or TIA were included even if the previous stroke occurred during ongoing OAC. The first breakthrough event that occurred during the study period served as the index stroke for ASPERA-R.

Eligibility required (1) age >18 years; (2) a documented diagnosis of AF; (3) any ischemic stroke (breakthrough event)—either first-ever or recurrent—diagnosed according to WHO criteria and confirmed by noncontrast CT (NCCT) and/or MRI; (4) occurrence during ongoing, continuous OAC therapy (either with a DOAC or vitamin K antagonist [VKA]) between February 2020 and February 2025; (5) availability of a baseline extracranial and intracranial brain vessel imaging (CT/MR angiography [CTA/MRA], color Doppler ultrasonography [CDU], or digital subtraction angiography [DSA]); and (6) availability of a 90 ± 10-day follow-up after the index stroke, obtained through an in-person visit with standardized outcome ascertainment.

Conversely, we excluded patients with (1) inadequate DOAC dosing, as defined by any regimen not meeting guideline criteria based on renal function, age, body weight, or drug interactions (including underdosing or unjustified dose reductions) and (2) poor compliance with anticoagulant therapy, defined as missed or uncertain OAC intake during the 7 days preceding the stroke based on patient/caregiver report or medical records.

To ensure reliable comparisons, we restricted the cohort to individuals with complete data for all predefined mandatory baseline variables. A full list of mandatory and optional baseline variables is provided in eAppendix 2.

Data Collection, Quality, and Study Definitions

Data were obtained through medical records during hospitalization and standard follow-up visits. To limit selection bias, all consecutive hospital admissions and emergency department

evaluations during the study period were screened and all eligible patients were included. Information was entered into a standardized Research Electronic Data Capture clinical research form. Data quality was ensured through weekly checks and central monitoring; queries were addressed to participating centers if needed. The analysis data set was locked on September 1, 2025.

Only centers with proven high-level stroke expertise and established, comprehensive diagnostic pathways were invited to join the ASPERA consortium; centers without a lead investigator with recognized stroke expertise or without a routinely exhaustive diagnostic workup were not included. Consequently, all participating centers applied a uniform and comprehensive diagnostic workflow for both the index breakthrough stroke and outcome events, including at minimum brain CT/MRI, vascular imaging (CTA, MRA, or duplex ultrasound), in-hospital cardiac rhythm monitoring, transthoracic echocardiography, and routine laboratory testing. Event adjudication was performed locally.

In addition, the etiologic assignment of the index event and the adjudication of outcome events were cross-checked centrally against all available study information. Whenever discrepancies or insufficient details were identified, centers were contacted for clarification to ensure consistency and accuracy across the data set.

Patients were categorized into 2 groups: those with and those without competing stroke etiologies. Competing stroke etiologies were defined as the presence of at least 1 additional plausible mechanism, beyond cardioembolism, which could have contributed to the index breakthrough stroke. Competing mechanisms were classified at the time of the index breakthrough stroke using the Trial of Org 10172 in Acute Stroke Treatment (TOAST) criteria.¹³ Competing etiologies were categorized as small vessel disease, large artery atherosclerosis (symptomatic stenosis ≥50% or occlusion of a relevant artery), or other determined causes. Large artery occlusions of extracranial vessels were attributed to large artery atherosclerosis only when vascular imaging demonstrated underlying atherosclerotic changes at the occlusion site (i.e., luminal irregularities, calcifications, and narrowing). In the absence of such features, occlusions were classified as cardioembolic. When ≥2 noncardioembolic mechanisms were present in addition to cardioembolism, we defined this as combined competing etiologies. Patients without competing etiologies were those in whom cardioembolism related to AF was the only identified mechanism contributing to the index stroke.

Study Outcomes

The primary outcome was the 90-day occurrence of a new ischemic stroke after the index breakthrough event, defined as a new neurologic deficit with a corresponding acute ischemic lesion on brain NCCT or MRI, including cases with symptom resolution within 24 hours (TIAs). Secondary outcomes

included the 90-day ordinal modified Rankin Scale (mRS) score shift, 90-day myocardial infarction, all-cause death, and vascular death (any death due to cardiovascular or cerebrovascular causes, including fatal ischemic or hemorrhagic stroke, sudden cardiac death, myocardial infarction, or other vascular events). Safety outcomes included 90-day moderate-to-severe bleeding, intracranial hemorrhage (ICH), 24-hour hemorrhagic transformation (HT), and 24-hour symptomatic ICH (sICH). Bleeding severity was classified according to the Global Utilization of Streptokinase and Tissue Plasminogen Activator for Occluded Coronary Artery trial classification.¹⁴ Specifically, we categorized the clinical impact of bleeding events as moderate to severe if they involved fatal hemorrhage or hemorrhage that, regardless of hemodynamic compromise, required blood or fluid replacement, inotropic support, or surgical intervention.¹⁴ HT was defined as any category according to the Heidelberg classification system.¹⁵ We defined sICH as any ICH accompanied by a ≥ 4 -point worsening on the NIH Stroke Scale (NIHSS) not attributable to other causes.¹⁶ All primary and secondary outcomes were ascertained by investigators through systematic review of 90-day follow-up visits, using predefined operational definitions.

Statistical Analysis

Categorical variables were summarized as counts (percentages) and continuous variables as mean (SD) or median (interquartile range [IQR]), as appropriate. Group comparisons used the χ^2 test (or Fisher exact test when expected counts < 5) for categorical variables; the Wilcoxon rank-sum test for continuous 2-group contrasts; and 1-way analysis of variance or the Kruskal-Wallis test for ≥ 3 groups, depending on distribution. Pairwise post hoc comparisons were adjusted using the Holm method.

To reduce baseline imbalances between patients with and without competing stroke etiologies, we applied inverse probability weighting (IPW).¹⁷ Propensity scores were estimated by logistic regression including prespecified covariates: age, ethnicity, enrolling center, cardiovascular risk factors, AF type, baseline NIHSS and mRS scores, and acute reperfusion therapies. Stabilized inverse weights were assigned to limit the influence of extreme values.

All outcome analyses were conducted in both unweighted and IP-weighted cohorts and were adjusted for prestroke covariates with imbalance (standardized mean difference [SMD] > 0.10). In all models, center-level clustering was accounted for using cluster-robust variance estimators. Time-to-event outcomes (90-day new ischemic stroke, myocardial infarction, all-cause and vascular death, moderate-to-severe bleeding, and ICH) were analyzed using weighted Cox models, reporting hazard ratios (HRs) with 95% confidence intervals (CIs). Kaplan-Meier curves illustrated survival, with group differences tested by log rank. Proportional hazards were assessed using Schoenfeld residuals; when violated, a 30-day Heaviside Cox model was applied to estimate early (0–30 days) and late (30–90 days) HRs. For all these outcomes, death was

considered a competing risk, as it could preclude the occurrence of the event. Cumulative incidence functions were compared using the Gray test, and subdistribution HRs (SHRs) were estimated using Fine-Gray regression. We further compared the risk of new ischemic stroke across competing etiology groups using modified Poisson regression with robust (sandwich) errors, with a global likelihood ratio test for overall differences. The 90-day mRS shift was analyzed using ordinal logistic regression (proportional odds), reporting adjusted odds ratios (aORs) and 95% CIs. 2HT and sICH within 24 hours were assessed using generalized linear models to obtain adjusted risk differences (aRDs). Finally, we performed a sensitivity analysis in the unweighted cohort restricted to patients without previous stroke or TIA (first-ever stroke) to compare outcomes between groups.

No formal sample-size calculation was performed; the cohort reflects all eligible patients enrolled during the study period. Analyses were conducted in R version 4.4.1 (R Foundation for Statistical Computing, Vienna, Austria), with 2-sided $p < 0.05$ considered statistically significant.

Data Availability

The complete data set used for this study will be shared on request from any qualified researcher to the corresponding author.

Results

A total of 1,649 patients (mean age of 78.0 ± 10.7 years, 47.8% male) were included in the analysis, of whom 24.3% had competing stroke etiologies. The study flowchart with the number of excluded patients is displayed in Figure 1. At least a brain neuroimaging examination was performed for all patients at baseline (brain NCCT 89.1%, brain MRI 7.0%, both 3.9%). Similarly, all patients underwent at least a brain vessel imaging examination (CTA 81.9%, MRA 10.9%, CDU 26.1%, DSA 44.6%), with no significant between-group differences in imaging modality distribution (SMDunw = 0.021).

Characteristics of Patients With Competing Stroke Etiologies

Large artery atherosclerosis was the most frequent competing etiology (240/401, 59.9%), followed by small vessel disease (70/401, 17.5%), other determined causes (60/401, 15.0%), and combined mechanisms (31/401, 7.7%). Symptomatic extracranial carotid stenosis $\geq 50\%$ was present in 155 of 401 patients (31.1%), vertebral stenosis in 30 of 155 (19.4%), and intracranial stenosis in 52 of 401 (13.0%), most often involving the middle cerebral artery (65.4%). Among “other determined etiologies,” cancer-related thrombophilia predominated (83.3%) (eTable 1).

In the unweighted cohort, patients with competing stroke etiologies were more often male (55.4% vs 45.4%; SMDunw = 0.200) and had greater stroke severity (median NIHSS score 12 [IQR 5–18] vs 8 [IQR 3–17]; SMDunw = 0.210). They

underwent endovascular thrombectomy less frequently (32.7% vs 48.2%; SMDunw = 0.320) and had a higher vascular risk burden, including hypertension (86.5% vs 79.4%; SMDunw = 0.190), dyslipidemia (58.6% vs 48.9%; SMDunw = 0.196), diabetes (32.2% vs 25.2%; SMDunw = 0.155), ischemic heart disease (30.7% vs 19.5%; SMDunw = 0.260), peripheral artery disease (8.7% vs 3.6%; SMDunw = 0.214), and previous stroke/TIA (30.9% vs 22.9%; SMDunw = 0.181). They were also more often on antiplatelet therapy at admission (13.7% vs 6.1%; SMDunw = 0.257) (Table 1).

Poststroke prevention also differed, with higher use of lipid-lowering drugs (73.3% vs 61.7%; SMDunw = 0.295), antiplatelet therapy (25.4% vs 11.9%; SMDunw = 0.376), and carotid revascularization (endarterectomy 4.2% vs 0.1%; stenting 2.2% vs 0.2%; SMDunw = 0.314). Most patients in both groups were prescribed DOACs (66.1% vs 75.1%), with no major differences in anticoagulant type or switching (SMDunw = 0.025). Notably, DOAC plus single antiplatelet therapy was far more common in competing etiology groups (24.4% vs 0.2%; SMDunw = 0.769) (Table 1).

Across competing etiology subtypes, stroke severity varied significantly (global $p < 0.001$), with higher NIHSS scores in large artery atherosclerosis vs small vessel disease and in other or combined etiologies vs small vessel disease. Diabetes was more frequent in patients with large artery atherosclerosis (global $p = 0.004$; adjusted $p = 0.013$ vs other etiologies). Antiplatelet use after the index stroke was most frequent in patients with large artery atherosclerosis (72/240, 30.0%): 60 received low-dose aspirin for symptomatic extracranial internal carotid artery stenosis $\geq 50\%$ while 10 were treated with clopidogrel and 2 with ticagrelor after recent percutaneous angioplasty with stent placement. Antiplatelets were also prescribed alongside anticoagulation in 13 patients with small vessel disease (18.6%; 11 aspirin, 2 clopidogrel), 12 with other determined causes (20.0%; all aspirin), and 5 with combined mechanisms (16.1%; 3 aspirin, 2 clopidogrel). Anticoagulation strategies also differed (global $p = 0.003$) without significant pairwise contrasts. Finally, carotid intervention (endarterectomy/stenting) was performed only in 26 of 104 patients (25.0%) with 50%–99% symptomatic extracranial internal carotid stenosis (Table 1, eTable 2).

Inverse Probability Weighting

IPW matched 806 patients with competing etiologies and 843 without. All variables used in the propensity model were complete, and balance diagnostics indicated excellent performance: the SMD of the propensity score was 0.021 (< 0.25), and the variance ratio was 1.00 (acceptable 0.5–2).¹⁷ Visual inspection of propensity scores' overlap between groups and covariate balance confirmed the adequacy of weighting (eFigures 1 and 2). The estimated overlap of propensity scores after weighting was good, with an estimated smooth overlap coefficient of 0.75 (good overlap > 0.70). Most weighted SMDs were < 0.10 across baseline characteristics (Table 1).

Primary Outcome

New ischemic stroke occurred in 26 patients (6.5%) with competing stroke etiologies and in 31 (2.5%) without (adjusted HR [aHR] 2.50, 95% CI 1.95–3.18; $p < 0.001$) (eTable 3). In the weighted cohort, the 90-day risk was 6.0% vs 2.7%, corresponding to an aHR of 2.62 (95% CI 2.01–3.42; $p < 0.001$) (Table 2, Figure 2). Competing risk analysis confirmed a significantly higher cumulative incidence in patients with competing etiologies (Gray test $\chi^2 = 10.25$, $p = 0.014$), and Fine-Gray regression yielded a SHR of 2.20 (95% CI 1.34–3.62; $p = 0.018$). No difference was observed for death as the competing event (Gray test $p = 0.254$) (eFigure 3).

Compared with patients without competing etiologies, the 90-day risk was higher for those with large artery atherosclerosis (risk ratio 2.20, 95% CI 1.30–3.71) and other determined causes (risk ratio 2.68, 95% CI 1.27–5.66). Small vessel disease showed no clear difference (risk ratio 1.19, 95% CI 0.37–3.90), and combined mechanisms suggested an increase not reaching significance (risk ratio 2.57, 95% CI 0.92–7.18) (Figure 3).

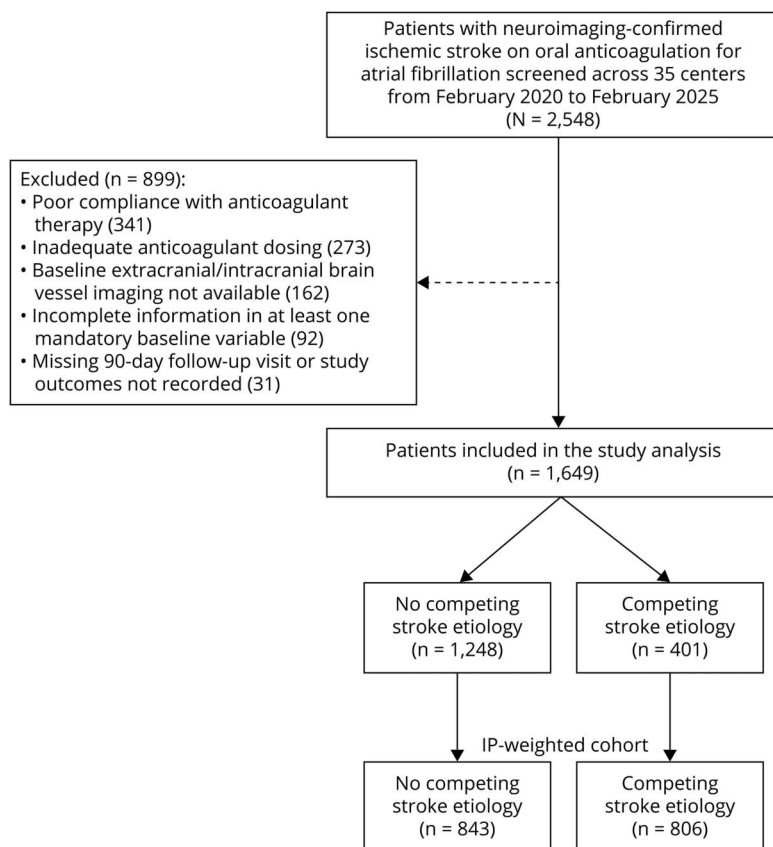
Secondary Outcomes

For the secondary outcomes, mRS scores were shifted toward greater disability in the competing etiology group both in the unweighted (aOR 1.24, 95% CI 1.03–1.49; $p = 0.023$) and weighted (aOR 1.30, 95% CI 1.08–1.55; $p = 0.005$) cohorts (Table 2, eTable 3, Figure 4). Among patients with recurrent ischemic stroke, the 90-day mRS distribution was also worse in those with competing etiologies (median 4, IQR 3–5 vs 3, IQR 2–4), yielding an aOR of 2.71 (95% CI 1.22–7.01; $p = 0.038$). Myocardial infarction did not differ (Table 2, eTable 3). Competing risk analysis showed a nonsignificant trend toward a higher incidence of the outcome in patients with competing etiologies (Gray test $\chi^2 = 3.04$, $p = 0.081$), and Fine-Gray regression yielded a SHR of 2.09 (95% CI 0.90–4.88; $p = 0.089$). No difference was observed for death as the competing event (Gray test $p = 0.183$) (eFigure 4). All-cause mortality was higher in the competing etiology group (aHR 1.34, 95% CI 1.05–1.72; $p = 0.011$, weighted aHR 1.58, 95% CI 1.18–2.12; $p = 0.002$), whereas vascular death did not differ (Table 2, eTable 3). For 90-day all-cause mortality, the proportional hazards assumption was violated (global Schoenfeld $p = 0.014$) and the log-rank test was not significant (0.137), suggesting a time-varying effect. The Heaviside Cox model showed no early difference (0–30 days: aHR 0.95, 95% CI 0.80–2.22; $p = 0.637$) but a late higher hazard (30–90 days: aHR 1.62, 95% CI 1.18–2.22; $p = 0.003$).

Safety Outcomes

For safety outcomes, patients with a competing stroke etiology had a higher 90-day risk of moderate-to-severe bleeding in both the unweighted (aHR 1.60, 95% CI 1.01–2.57; $p = 0.048$) and weighted (aHR 1.82, 95% CI 1.08–3.09; $p = 0.026$) cohorts (Table 2, eTable 3). Proportional hazards were violated (global Schoenfeld $p < 0.001$), but the log-rank test remained significant

Figure 1 Flowchart of the ASPERA-R Study



ASPERA = Advancing Knowledge in Ischemic Stroke Patients on Oral Anticoagulants; IP = inverse probability.

($p = 0.006$). In the Heaviside Cox model, no early difference was observed (0–30 days: aHR 1.00, 95% CI 0.37–2.72; $p = 0.994$), whereas risk was higher between 30 and 90 days (aHR 2.42, 95% CI 1.11–5.31; $p = 0.026$) (Table 2, Figure 5). Competing risk analysis showed a significantly higher cumulative incidence of the event among patients with competing etiologies (Gray test $\chi^2 = 7.56$, $p = 0.006$), consistent with the Fine-Gray model, which yielded a SHR of 2.03 (95% CI 1.21–3.40; $p = 0.007$). No difference was observed for death as the competing event (Gray test $p = 0.499$) (eFigure 5). Among extracranial moderate-to-severe events in patients with competing etiologies ($n = 16/36$, 44.4%), most were gastrointestinal (56.2%), followed by soft-tissue hematoma (25.0%), genitourinary (12.5%), and respiratory (6.3%) sites.

ICH did not differ (Table 2, eTable 3, and eFigure 6), while HT at 24 hours was less frequent in the competing etiology group (unweighted aRD –6.7%, 95% CI –11.1% to –2.4%; $p = 0.002$; weighted aRD –4.3%, 95% CI –7.8% to –0.7%; $p = 0.020$). sICH was not significant after IPW (aRD 0.8%, 95% CI –2.9% to 0.3%; $p = 0.119$) (Table 2, eTable 3).

Sensitivity Analysis

Our sensitivity analysis in the unweighted cohort, restricted to patients with first-ever stroke, confirmed the findings of the

primary analysis for most outcomes, including 90-day recurrent ischemic stroke, 90-day all-cause death, and 24-hour hemorrhagic transformation. In addition, it revealed a consistent trend toward a worse functional outcome distribution and a higher incidence of moderate-to-severe bleeding among patients with competing etiologies (eTable 4).

Discussion

In this large multicenter cohort of AF patients with breakthrough ischemic stroke despite continuous OAC, nearly one-quarter had competing etiologies in addition to cardioembolism. This was associated with higher risks of recurrent ischemic stroke, poorer functional recovery, increased mortality, and greater bleeding. These findings highlight the clinical complexity of breakthrough strokes on OAC, where both index and recurrent events may result from the coexistence of AF-related cardioembolism and other mechanisms. The ASPERA-R study extends previous evidence by offering a detailed characterization of risk factors and clinical features and by assessing a broad spectrum of outcomes in a large, rigorously curated cohort, thereby providing a more comprehensive understanding of the prognostic impact of competing etiologies.

Table 1 Baseline Characteristics

	Overall cohort (n = 1,649)	Competing stroke etiology (n = 401)	No competing stroke etiology (n = 1,248)	SMD unweighted	SMD weighted
Demographics					
Age, y, mean ± SD	78.9 ± 10.60	78.3 ± 9.68	78.4 ± 10.60	0.005	0.083
Sex, male, n (%)	789 (47.8)	222 (55.4)	567 (45.4)	0.200	0.040
Ethnicity, n (%)				0.136	0.068
Non-Hispanic White	1,303 (79.0)	312 (77.8)	991 (79.4)		
Hispanic White	113 (6.9)	38 (9.5)	75 (6.0)		
Black	24 (1.5)	5 (1.2)	29 (1.5)		
Asian	9 (0.5)	2 (0.5)	7 (0.6)		
Other	200 (12.1)	44 (11.0)	156 (12.5)		
Baseline oral anticoagulation characteristics					
Type of oral anticoagulation at the time of index stroke, n (%)				0.079	0.022
VKA	374 (22.7)	81 (20.2)	293 (23.5)		
DOAC	1,275 (77.3)	320 (79.8)	955 (76.5)		
Type of DOAC at the time of index stroke, n (%)				0.077	0.067
Apixaban	469/1,275 (36.8)	127/320 (39.7)	342/955 (35.8)		
Rivaroxaban	428/1,275 (33.6)	104/320 (32.5)	324/955 (33.9)		
Edoxaban	251/1,275 (19.7)	60/320 (18.8)	191/955 (20.0)		
Dabigatran	127/1,275 (10.0)	29/320 (9.1)	98/955 (10.3)		
Time from last DOAC intake to admission, n (%)					
<12 h	653/1,275 (51.2)	164/320 (51.2)	489/955 (51.2)	0.012	0.014
12–24 h	471/1,275 (36.9)	116/320 (36.2)	355/955 (37.2)		
24–48 h	151/1,275 (11.9)	40/320 (12.5)	111/955 (11.6)		
INR on admission, mean ± SD	1.39 ± 0.54	1.38 ± 0.49	1.40 ± 0.56	0.032	0.011
DOAC levels on admission available, n (%)	281/1,275 (22.0)	82 (25.6)	199 (20.8)	0.113	0.093
DOAC levels on admission, n (%)				0.121	0.005
Below range	67/281 (23.9)	16/82 (19.5)	51/199 (25.6)		
Within range	197/281 (70.1)	61/82 (74.4)	136/199 (68.3)		
Above range	17/281 (6.0)	5/82 (6.1)	12/199 (6.0)		
Clinical characteristics					
Hospitalization, n (%)	1,596 (96.8)	391 (97.5)	1,205 (96.6)	0.056	0.002
Hospital setting, n (%)					
Stroke unit	1,504/1,596 (94.2)	371/391 (94.9)	1,133/1,205 (94.0)	0.054	0.096
Intensive care unit	42/1,596 (2.6)	11/391 (2.8)	31/1,205 (2.6)		
Other hospital units	50/1,596 (3.1)	9/391 (2.3)	41/1,205 (3.4)		
NIHSS score on admission, median (IQR)	11 (5–18)	12 (5–18)	8 (3–17)	0.210	0.082
Prestroke mRS score category, n (%)				0.196	0.122 ¹

Continued

Table 1 Baseline Characteristics (*continued*)

	Overall cohort (n = 1,649)	Competing stroke etiology (n = 401)	No competing stroke etiology (n = 1,248)	SMD unweighted	SMD weighted
No symptoms (score of 0)	814 (49.4)	173 (43.1)	641 (51.4)		
Symptoms without any disability (score of 1)	369 (22.4)	93 (23.2)	276 (22.1)		
Symptoms with mild disability (score of 2)	223 (13.5)	62 (15.5)	161 (12.9)		
Symptoms with mild-to-moderate disability (score of 3)	179 (10.9)	54 (13.5)	125 (10.0)		
Symptoms with moderate-to-severe disability (score of 4)	59 (3.6)	16 (4.0)	43 (3.4)		
Symptoms with severe disability (score of 5)	5 (0.3)	3 (0.7)	2 (0.2)		
IV thrombolysis, n (%)	139 (8.4)	28 (7.0)	111 (8.9)	0.071	0.023
Endovascular thrombectomy, n (%)	732 (44.4)	131 (32.7)	601 (48.2)	0.320	0.062
Risk factors					
Arterial hypertension, n (%) ^a	1,338 (81.1)	347 (86.5)	991 (79.4)	0.190	0.065
Dyslipidemia, n (%) ^b	845 (51.2)	235 (58.6)	610 (48.9)	0.196	0.012
Diabetes, n (%) ^c	443 (26.9)	129 (32.2)	314 (25.2)	0.155	0.055
Cigarette smoking, n (%) ^d	154 (9.3)	47 (11.7)	107 (8.6)	0.104	0.074
Previous ischemic stroke or TIA, n (%)	410 (24.9)	124 (30.9)	286 (22.9)	0.181	0.151 ⁱ
Previous intracranial hemorrhage, n (%)	27 (1.6)	8 (0.9)	19 (2.4)	0.060	0.095
Ischemic heart disease, n (%) ^e	366 (22.2)	123 (30.7)	243 (19.5)	0.260	0.171 ⁱ
Chronic congestive heart failure, n (%) ^f	277 (16.8)	77 (19.2)	200 (20.0)	0.083	0.053
Chronic kidney disease, n (%) ^g	268 (16.3)	78 (19.5)	190 (15.2)	0.112	0.053
Chronic liver failure, n (%) ^h	7 (0.4)	3 (0.7)	4 (0.3)	0.059	0.096
Symptomatic peripheral artery disease, n (%) ⁱ	80 (4.9)	35 (8.7)	45 (3.6)	0.214	0.075
Mechanical heart valve, n (%)	94 (5.7)	20 (5.0)	74 (5.9)	0.041	0.027
Biological heart valve, n (%)	81 (4.9)	22 (5.5)	59 (4.7)	0.034	0.032
Atrial fibrillation type, n (%) ^j				0.302	0.198 ⁱ
Paroxysmal	335 (20.3)	109 (27.2)	226 (18.1)		
Persistent	200 (12.1)	47 (11.7)	153 (12.3)		
Long-standing persistent	76 (4.6)	16 (4.0)	60 (4.8)		
Permanent	847 (51.4)	204 (50.9)	643 (51.5)		
Unknown	191 (11.6)	25 (6.2)	116 (13.3)		
Antihypertensive drugs on admission, n (%)	1,310 (79.4)	330 (82.3)	980 (78.5)	0.095	0.059
Lipid-lowering drugs on admission, n (%)	742 (45.0)	205 (51.1)	537 (43.0)	0.163	0.097
Antidiabetic drugs on admission, n (%)	397 (24.1)	112 (27.9)	285 (22.8)	0.117	0.045
Rhythm control drugs on admission, n (%)	1,088 (66.0)	272 (67.8)	816 (65.4)	0.052	0.074
Antiplatelet therapy on admission, n (%)	131 (7.9)	55 (13.7)	76 (6.1)	0.257	0.221 ⁱ
Poststroke secondary prevention					
Poststroke anticoagulation strategy, n (%)				0.025	0.001

Continued

Table 1 Baseline Characteristics (continued)

	Overall cohort (n = 1,649)	Competing stroke etiology (n = 401)	No competing stroke etiology (n = 1,248)	SMD unweighted	SMD weighted
No anticoagulation restarted ^k	179 (10.9)	33 (8.2)	146 (11.7)		
Same DOAC restarted	463 (28.1)	127 (31.7)	336 (26.9)		
Switch to different DOACs (anti-Xa ↔ anti-Xa/anti-IIa ↔ anti-IIa)	201 (12.2)	45 (11.2)	156 (12.5)		
Switch to different DOACs (anti-Xa ↔ anti-IIa)	283 (17.2)	70 (17.5)	213 (17.1)		
VKA → DOAC	112 (6.8)	23 (5.7)	89 (7.1)		
VKA → VKA	179 (10.9)	40 (10.0)	139 (11.1)		
DOAC → VKA	59 (3.6)	12 (3.0)	47 (3.7)		
DOAC → LMWH	36 (2.2)	14 (3.5)	22 (1.8)		
VKA → LMWH	19 (1.2)	3 (0.7)	16 (1.3)		
Left atrial appendage closure	8 (0.5)	0 (0.0)	8 (0.6)		
Unknown	110 (6.7)	34 (8.5)	76 (6.1)		
Days from index stroke to anticoagulation restart, mean ± SD	8.1 ± 9.1	7.8 ± 9.2	8.2 ± 9.0	0.047	0.017
Anticoagulation discontinued during follow-up, n (%)	151/1,350 (11.2)	53/334 (15.9)	98/1,026 (9.6)	0.038	0.004
Percentage of follow-up time on anticoagulation, mean ± SD	91.1 ± 10.1	91.5 ± 10.3	91.0 ± 10.1	0.049	0.022
Poststroke antihypertensive drugs, n (%)	1,346 (81.7)	341 (85.0)	1,005 (80.5)	0.120	0.176 ^l
Poststroke lipid-lowering drugs, n (%)	1,064 (64.5)	294 (73.3)	770 (61.7)	0.295	0.294 ^l
Poststroke antidiabetic drugs, n (%)	535 (32.4)	115 (28.7)	272 (21.8)	0.169	0.137 ^l
Poststroke rhythm control drugs, n (%)	1,147 (69.6)	735 (58.9)	239 (59.6)	0.037	0.093
Poststroke antiplatelet therapy, n (%)	250 (15.2)	102 (25.4)	148 (11.9)	0.376	0.387 ^l
Poststroke carotid intervention, n (%)				0.314	0.351 ^l
Endarterectomy	18 (1.1)	17 (4.2)	1 (0.1)		
Stenting	12 (0.7)	9 (2.2)	3 (0.2)		

Abbreviations: DOAC = direct oral anticoagulant; INR = international normalized ratio; IQR = interquartile range; mRS = modified Rankin Scale; NIHSS = NIH Stroke Scale; SMD = standardized mean difference; VKA = vitamin K antagonist.

^a Arterial hypertension was defined as a history of blood pressure >140/90 mm Hg or the current use of antihypertensive medications.

^b Dyslipidemia was defined as history of total blood cholesterol levels >220 mg/dL and/or total triglycerides levels >130 mg/dL and/or the current use of lipid-lowering drugs.

^c Diabetes mellitus was defined as history of fasting glucose >126 mg/dL or the current use of hypoglycemic medications.

^d Current smoking was defined as the consumption of ≥1 cigarette per day over the past year.

^e Ischemic heart disease was defined as history of myocardial infarction, angina, or previous evidence of coronary disease on coronary angiography.

^f Chronic congestive heart failure was defined as history of stage C (structural heart disease and current or past history of heart failure symptoms) or stage D (refractory symptoms that interfere with daily life or recurrent hospitalization despite targeted guideline-directed medical therapy) chronic heart failure.

^g Chronic kidney disease was defined as history of estimated creatinine clearance of less than 60 for 3 months or more (including dialysis).

^h Chronic liver failure was defined as history of cirrhosis or end-stage liver disease.

ⁱ Symptomatic atherosclerotic peripheral artery disease was defined as history of intermittent claudication of presumed atherosclerotic origin.

^j Atrial fibrillation type was classified according to the American College of Cardiology/American Heart Association/Heart Rhythm Society guidelines: paroxysmal atrial fibrillation refers to patients with only episodes that terminate spontaneously within 7 days; persistent atrial fibrillation lasts more than 7 days or requires cardioversion to restore sinus rhythm; long-standing persistent atrial fibrillation is a continuous arrhythmia lasting more than 12 months with an ongoing rhythm-control intent; permanent atrial fibrillation refers to patients in whom the arrhythmia is accepted and no further attempts to restore sinus rhythm are pursued.

^k 121 of 179 (67.6%) died during the hospital stay; 16 of 179 patients (8.9%) had severe in-hospital bleeding events, mainly ICH (n = 11/16); 31 of 179 (17.3%) had no identifiable clinical reason for withholding OAC after the index stroke.

^l SMD >0.10

Table 2 Outcomes Comparison in the Weighted Cohort

	Weighted cohort		Statistical metric ^a	Weighted difference (95% CI)	<i>p</i> Value
	Competing stroke etiology (n = 806)	No competing stroke etiology (n = 843)			
Primary outcome					
90-d new ischemic stroke, n (%)	48 (6.0)	23 (2.7)	Adjusted hazard ratio	2.62 (2.01 to 3.42)	<0.001 ^b
Secondary outcomes					
90-d mRS score distribution, n (%)			Adjusted odds ratio	1.30 (1.08 to 1.55)	0.005 ^b
No symptoms (score of 0)	83 (10.3)	107 (12.7)			
Symptoms without any disability (score of 1)	125 (15.5)	145 (17.2)			
Symptoms with mild disability (score of 2)	133 (16.5)	127 (15.1)			
Symptoms with mild-to-moderate disability (score of 3)	121 (15.0)	128 (15.2)			
Symptoms with moderate-to-severe disability (score of 4)	120 (14.9)	117 (13.9)			
Symptoms with severe disability (score of 5)	45 (5.6)	59 (7.0)			
Death (score of 6)	179 (22.2)	160 (19.0)			
90-d myocardial infarction, n (%)	16 (2.0)	8 (0.9)	Adjusted hazard ratio	2.12 (0.82 to 5.46)	0.119
90-d all-cause death, n (%)	179 (22.2)	160 (19.0)	Adjusted hazard ratio	1.58 (1.18 to 2.12)	0.002 ^b
90-d vascular death, n (%)	108 (13.4)	107 (12.7)	Adjusted hazard ratio	1.21 (0.81 to 1.82)	0.349
Safety outcomes					
90-d moderate-to-severe bleeding, n (%)	42 (5.2)	22 (2.6)	Adjusted hazard ratio	1.82 (1.08 to 3.09)	0.026 ^b
90-d intracranial hemorrhage, n (%)	10 (1.2)	13 (1.5)	Adjusted hazard ratio	0.56 (0.31 to 1.01)	0.057
24-h hemorrhagic transformation, n (%)	114 (14.1)	156 (18.5)	Adjusted risk difference (%)	−4.3 (−7.8 to −0.7)	0.020 ^b
24-h symptomatic intracranial hemorrhage, n (%)	16 (2.0)	29 (3.4)	Adjusted risk difference (%)	0.8 (−2.9 to 0.3)	0.119

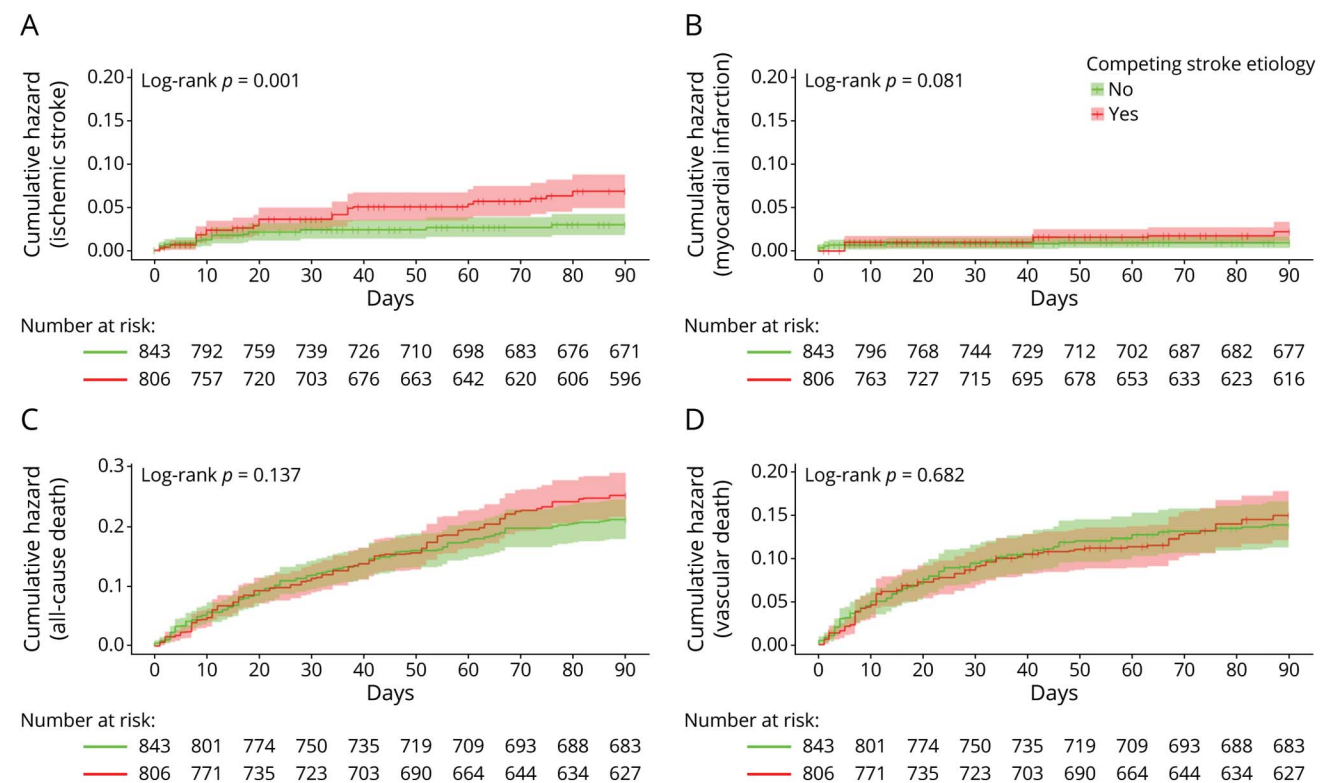
Global Schoenfeld residuals and *p* values: 90-day new ischemic stroke = 0.331; 90-day myocardial infarction = 0.162; 90-day vascular death = 0.014; 90-day all-cause death = 0.071; 90-day moderate-to-severe bleeding = <0.001; 90-day intracranial hemorrhage = 0.094.

^a Adjusted analyses controlled for prestroke variables that remained imbalanced after weighting (standardized mean difference >0.10), specifically, prestroke modified Rankin Scale scores, previous ischemic stroke or TIA, ischemic heart disease, atrial fibrillation type, and antiplatelet therapy at admission and incorporated center-level clustering using robust variance estimators.

^b *p* Values are statistically significant.

In our cohort, 24.3% of patients had at least 1 additional mechanism beyond cardioembolism contributing to the index breakthrough stroke, most commonly large artery atherosclerosis. Compared with those without competing causes, these patients presented with more severe strokes, a heavier burden of vascular risk factors, extracerebral atherosclerotic disease, and more frequent antiplatelet use. While prevalence was consistent with previous reports,^{11,18,19} we observed that competing etiologies were linked to higher admission NIHSS scores and greater cardiovascular

comorbidity, highlighting a clinically meaningful high-risk profile. Subtype analyses showed that large artery atherosclerosis was associated with greater severity and diabetes clustering, whereas small vessel disease tended to be milder. Anticoagulation strategies varied without a dominant pattern. Overall, these findings emphasize the heterogeneity of competing etiologies and support a tailored, mechanism-informed approach that extends beyond anticoagulation to address coexisting atherosclerotic disease and other cause-specific risk factors.

Figure 2 Cumulative Hazard Curves for Outcome Comparison in the Weighted Cohort

(A) 90-day new ischemic stroke, (B) 90-day myocardial infarction, (C) 90-day all-cause death, and (D) 90-day vascular death. Dashed areas indicate 95% CIs.

Competing etiologies were not only common but also associated with a substantially higher risk of recurrent ischemic stroke. Specifically, we found that patients with competing stroke mechanisms had a twofold higher risk of recurrence, which is higher than reported in the largest available study (aOR 1.83).¹⁸ This may in part reflect differences in secondary prevention patterns. Notably, in a large multinational

retrospective study,¹⁸ the widespread use of DOACs after the index stroke (>80%) was associated with a comparatively lower recurrence rate, suggesting a protective effect compared with other secondary prevention approaches. In our cohort, by contrast, only 66.1% of patients were prescribed DOACs after the index event, which may partially account for the higher risk we observed.

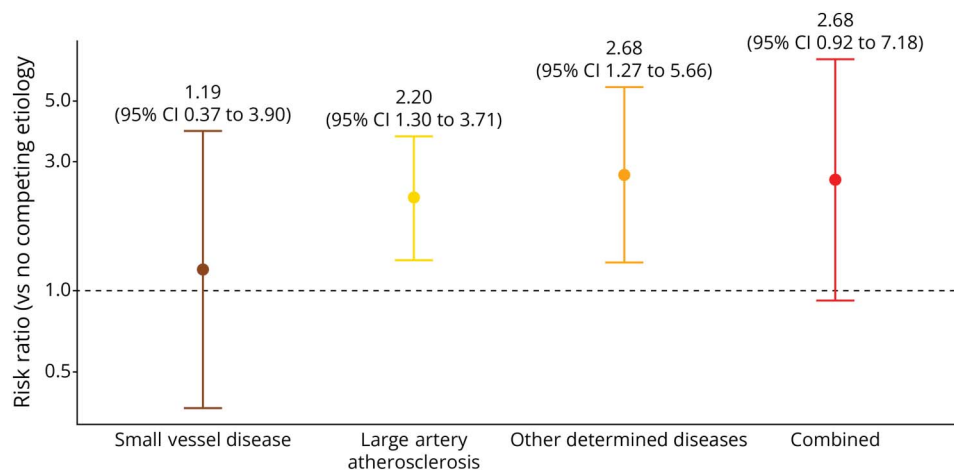
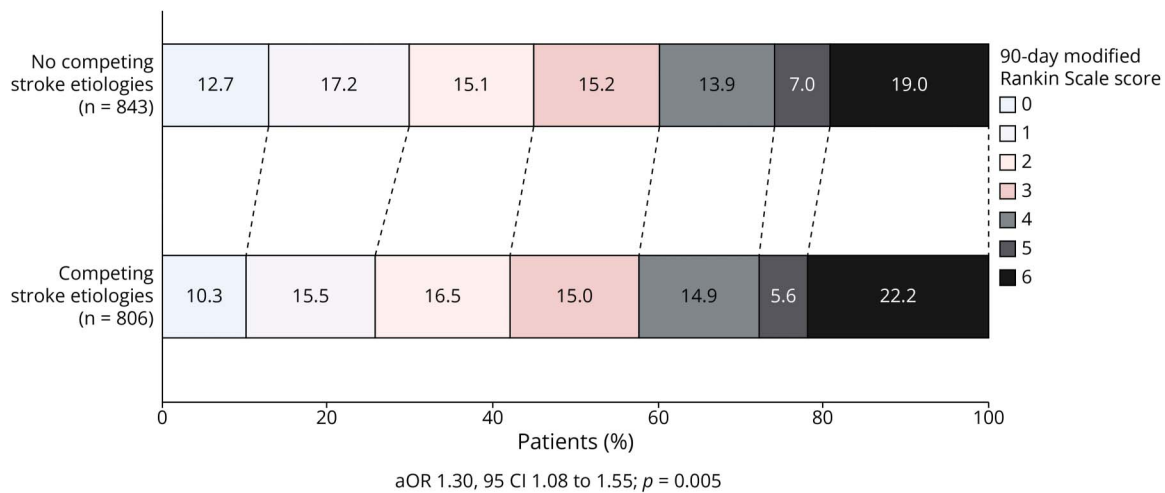
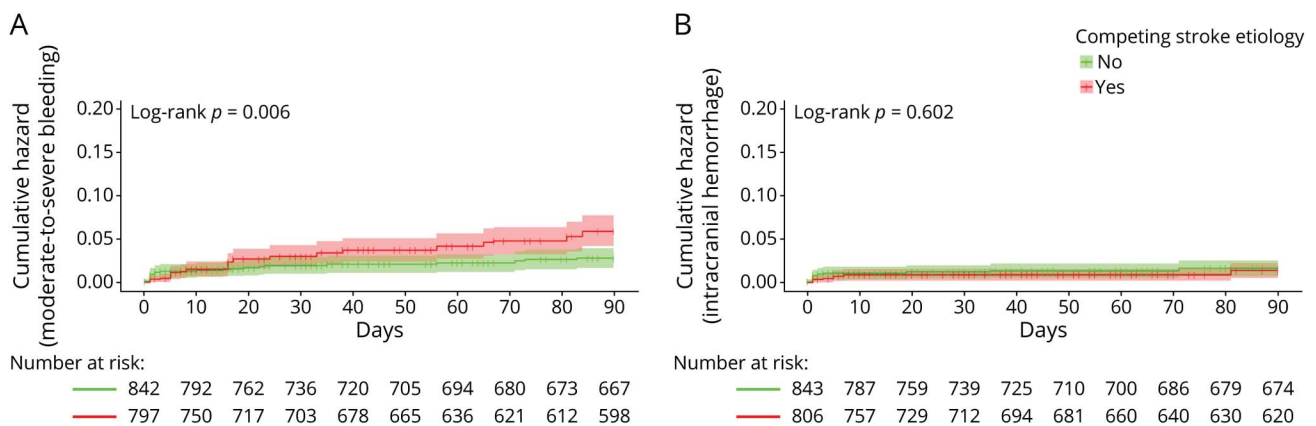
Figure 3 Risk of 90-Day Ischemic Stroke Stratified by Competing Etiology in the Weighted Cohort

Figure 4 90-Day mRS Score Shift in the Weighted Cohort

aOR = adjusted odds ratio.

Regarding specific competing mechanisms, large artery atherosclerosis, predominantly driven by symptomatic internal carotid artery stenosis $\geq 50\%$, showed the strongest association with ischemic stroke recurrence in our cohort. Overall, the presence of large artery atherosclerosis was linked to more than a twofold higher risk of new ischemic stroke compared with patients with AF alone, consistent with previous evidence.^{11,18} Of interest, we observed a substantially high risk of new ischemic stroke also in patients with other determined causes, most commonly cancer-related thrombophilia. Malignancy is known to consistently increase the risk of ischemic recurrence, with rates reported as high as 9.9% at 1 month and 25% at 6 months in patients with cryptogenic stroke.^{20,21} These findings have important clinical implications, as further diagnostic workup may be warranted in selected patients with breakthrough stroke on OAC for AF to uncover occult malignancy.

For the secondary outcomes, we observed a significantly worse 90-day mRS score shift in patients with competing etiologies compared with those without; in the weighted cohort, 57.7% had mRS score ≥ 3 at 90 days. This seems to be a novel observation and is plausibly driven by a combination of greater index stroke severity, higher rates of ischemic recurrence, and a heavier vascular risk factor burden, all of which constrain neurologic recovery. It is important to note that recurrent events in this subgroup were also more disabling within our cohort, amplifying the risk of long-term dependency. We also observed higher mortality rates among patients with competing etiologies compared with those without. Previous evidence showed that the coexistence of large artery atherosclerosis and cardioembolism confers an excess risk of death,²² which may explain our finding, given that large artery atherosclerosis was the most prevalent competing etiology in our cohort. Notably,

Figure 5 Cumulative Hazard Curves for Comparison of Safety Outcomes in the Weighted Cohort

(A) 90-day moderate-to-severe bleeding and (B) 90-day intracranial hemorrhage. Dashed areas indicate 95% CIs.

this excess risk became evident over the longer term (between days 30 and 90), underscoring the need for sustained vigilance.

For safety outcomes, patients with competing etiologies had a higher 90-day risk of moderate-to-severe bleeding while ICH rates were comparable between groups. Time-varying analyses showed that this excess bleeding risk emerged after the first month, likely reflecting frequent use of combined low-dose aspirin in patients with concomitant atherosclerosis, a strategy previously shown to increase bleeding without reducing ischemic recurrence.^{18,23} Clinically, these findings emphasize the need for individualized antithrombotic regimens, with regular reassessment of bleeding risk and targeted risk mitigation measures. On the contrary, the low rate of carotid revascularization in our cohort probably reflects real-world clinical decision making, in which breakthrough strokes are predominantly managed as purely cardioembolic events and revascularization is often deferred in older, comorbid patients, even when moderate-to-severe carotid stenosis is present. Of interest, patients with competing etiologies had a lower risk of HT, plausibly reflecting the predominance of “pure” cardioembolic strokes in the non-competing group, as these are typically associated with larger infarcts, reperfusion dynamics, and greater intrinsic HT risk.^{24,25}

The principal strength of this study lies in the rigorous procedures adopted to ensure data accuracy and quality, reinforced by systematic checks of the ASPERA-R electronic database. Nevertheless, several limitations must be acknowledged. First, although we applied IPW and additional covariate adjustment, the observational design cannot fully exclude residual confounding. Second, the retrospective nature of the cohort entails risks of information bias. Despite systematic screening, some breakthrough strokes on OAC may have been missed, particularly in patients managed outside stroke units, limiting causal inference. Third, stroke mechanisms were classified using TOAST, which, although widely used, may oversimplify causal complexity and underestimate overlapping etiologies compared with ASCOD.²⁶ Moreover, index strokes without identified competing causes were pragmatically attributed to AF-related cardioembolism, which may have led to misclassification in patients with unrecognized competing mechanisms. Occult malignancies may have been underdetected because of lack of systematic cancer screening and the short 90-day follow-up. Fourth, we did not collect information on whether patients with a previous stroke or TIA experienced that event while already on OAC, nor on its underlying etiology. Although these cases represented less than one-quarter of our cohort (24.9%) and a competing cause was likely assessed clinically, the preceding therapeutic pathway was not documented. This may have limited our ability to fully characterize long-term etiologic trajectories. Anyway, our sensitivity analysis restricted to patients with first-ever ischemic stroke yielded results that aligned with our primary analysis. Fifth, we defined compliance using pragmatic exclusion criteria based on documented OAC intake rather than laboratory surrogates. DOAC plasma concentrations were available in only approximately 20% of patients, and time in therapeutic international normalized ratio (INR) was not documented for those treated

with VKAs, potentially leading to misclassification of on-treatment exposure. Sixth, patients with missing mandatory baseline or 90-day follow-up data ($n = 123$) were excluded. Although selection bias cannot be excluded, these patients did not differ from those included in terms of age (79.4 ± 10.1 vs 78.9 ± 10.6 years, $p = 0.680$) or index stroke severity (NIHSS score 12 [5–18] vs 11 [5–18], $p = 0.742$), indicating that their exclusion is unlikely to have affected cohort representativeness. Finally, most patients were recruited in Italy (60%), resulting in a predominance of non-Hispanic White individuals. This geographic concentration may limit generalizability and leave residual country-level confounding.

In conclusion, competing stroke mechanisms coexisting with cardioembolism are common in AF patients with breakthrough stroke and are linked to higher risks of recurrent stroke, bleeding, and disability. These findings highlight the importance of a systematic diagnostic workup beyond standard vascular imaging. Optimizing secondary prevention should extend beyond anticoagulation strategies to include rigorous management of cardiovascular comorbidities, such as statin use, blood pressure control, and other risk factor modifications. Finally, it remains essential to clarify whether specific competing etiology subtypes may benefit from the addition of antiplatelet therapy to anticoagulation.

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